

Arithmetic Progressions (Set B) — MCQ Worksheet

Mathematics • Chapter 5 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Common difference of AP: 11, 8, 5, 2, ...

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|--------|---------|
| (A) 3 | (B) -3 |
| (C) 11 | (D) -11 |

Q2. First term of AP whose 4th term is 22 and common difference is 5:

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|-------|--------|
| (A) 3 | (B) 5 |
| (C) 7 | (D) 12 |

Q3. n-th term of AP 3, 8, 13, 18, ... is:

- | | |
|--------------|--------------|
| (A) $5n + 3$ | (B) $5n - 2$ |
| (C) $3n + 5$ | (D) $5n$ |

Q4. 15th term of AP 21, 18, 15, ... is:

- | | |
|---------|---------|
| (A) -18 | (B) -21 |
| (C) 21 | (D) -24 |

Q5. Sum of first 20 terms of AP 2, 5, 8, ... is:

- | | |
|---------|---------|
| (A) 590 | (B) 610 |
| (C) 620 | (D) 650 |

Q6. If the nth term of AP is $4n - 1$, then common difference is:

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|--------|-------|
| (A) 3 | (B) 4 |
| (C) -1 | (D) 1 |

Q7. Number of terms in AP 5, 9, 13, ..., 81 is:

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|--------|--------|
| (A) 18 | (B) 19 |
| (C) 20 | (D) 21 |

Q8. Sum of first 10 terms of AP -5, -2, 1, ... is:

- | | |
|--------|---------|
| (A) 85 | (B) -85 |
| (C) 95 | (D) -95 |

Q9. If $a = 2$ and $d = 5$, then $a_{10} =$

- | | |
|--------|--------|
| (A) 45 | (B) 47 |
| (C) 50 | (D) 52 |

Q10. Three consecutive terms of an AP whose sum is 24 and product is 384 (assume smallest as $a-d$, a , $a+d$):

- | | |
|--------------|--------------|
| (A) 4, 8, 12 | (B) 6, 8, 10 |
| (C) 2, 8, 14 | (D) 8, 8, 8 |

Q11. Which term of AP 4, 9, 14, 19, ... is 124?

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|----------|----------|
| (A) 23rd | (B) 24th |
| (C) 25th | (D) 26th |

Q12. Sum of all natural numbers between 200 and 300 divisible by 4 is:

- | | |
|----------|----------|
| (A) 6200 | (B) 6300 |
| (C) 6400 | (D) 6500 |

Q13. If a , b , c are in AP, then $2b =$

- | | |
|-------------|-------------|
| (A) $a + c$ | (B) $a - c$ |
| (C) ac | (D) a/c |

Q14. 11th term of AP whose first term = -5 and last term = 45 with 11 terms total is:

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|--------|--------|
| (A) 45 | (B) 40 |
| (C) 35 | (D) 30 |

Q15. If 7 times the 7th term equals 11 times the 11th term, then the 18th term equals:

- (A) 0 (B) 1
(C) 18 (D) -18

Q16. How many three-digit numbers are divisible by 7?

- (A) 128 (B) 127
(C) 130 (D) 126

Q17. The sum of first n terms of an AP is $3n^2 + 5n$. Find the 25th term.

- (A) 152 (B) 155
(C) 158 (D) 160

Q18. If the ratio of the sum of m terms of an AP to the sum of n terms is $m^2 : n^2$, then ratio of m -th and n -th term is:

- (A) $(2m - 1) : (2n - 1)$ (B) $m : n$
(C) $m^2 : n^2$ (D) $m : 2n$

Q19. A man saves Rs 200 in first month, Rs 250 in second, Rs 300 in third and so on. His total saving in 2 years is:

- (A) Rs 11,400 (B) Rs 12,200
(C) Rs 12,600 (D) Rs 13,000

Q20. If the angles of a quadrilateral are in AP and smallest is 60° , then the largest is:

- (A) 120° (B) 150°
(C) 100° (D) 140°

Q21. If S_n denotes sum of first n terms of an AP, then $S_{\{30\}} - 3(S_{\{20\}} - S_{\{10\}}) =$

- (A) 0 (B) $S_{\{30\}}$
(C) $S_{\{10\}}$ (D) $-S_{\{10\}}$

Q22. In an AP, the sum of three numbers is 12 and sum of their cubes is 288. The numbers are:

- (A) 2, 4, 6 (B) 1, 4, 7
(C) 3, 4, 5 (D) 0, 4, 8

Q23. The sum of first 100 terms of the AP 1, 4, 7, 10, ... is:

- (A) 14950 (B) 14850
(C) 15050 (D) 15150

Q24. Find the AP whose 5th term is 19 and 13th term is 27 less than 18th term.

- (A) First term 3, $d = 4$ (B) First term 7, $d = 3$
(C) First term 5, $d = 4$ (D) Cannot be determined uniquely

Q25. If 5 times the 5th term of an AP is equal to 8 times its 8th term, find its 13th term.

- (A) 0 (B) 1
(C) 5 (D) 8